

5 Results

5.1 Validation on the Classical Puzzle

We checked our implementation by comparing it to the classic Blue-Eyed Islander puzzle where the agents have full visibility. We can use these result to get a good reference point so we know what outcomes to expect. Figure 10 shows that the number of days before convergence equals the number of blue-eyed people and therefore demonstrates that the simulation correctly captures the iterative reasoning process that is intended and not converges too early on account of an error from the way the updates are performed.

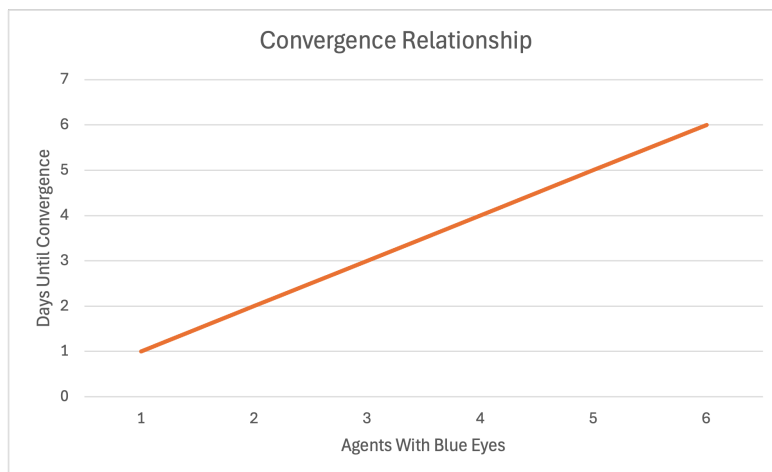


Figure 10: Days it takes to converge for each amount of agents with blue eyes when there is full visibility.

In Figure 11 you can see the case where everyone's eyes are blue and that the number of candidate world-states is eliminated in the typical step-wise fashion over the course of the days until only one possible world-state remains. This is the behavior that would be expected if the mechanism for public updates eliminates the correct candidate world-states after every observed day. With 6 agents, you start with 64 possible worlds (2^6). On Day 0 you can eliminate the worlds where everyone is brown because of the public announcement that says there is at least one blue-eyed agent. Then on Day 1 you can eliminate the 6 worlds where there is only one agent with blue eyes, as no one left on day 0. On Day 2 you can remove the 15 worlds that are shown in Figure 11 where there are two blue-eyed agents, as no one left on Day 1. This correctly continues until there is only one world left.

Simple edge cases were also checked to make sure they work correctly. If there is a single agent, it will always converge as soon as it hears the initial announcement, but the case with no visibility will never converge because there is not sufficient information generated through the public observation of the environment to eliminate the ambiguity.

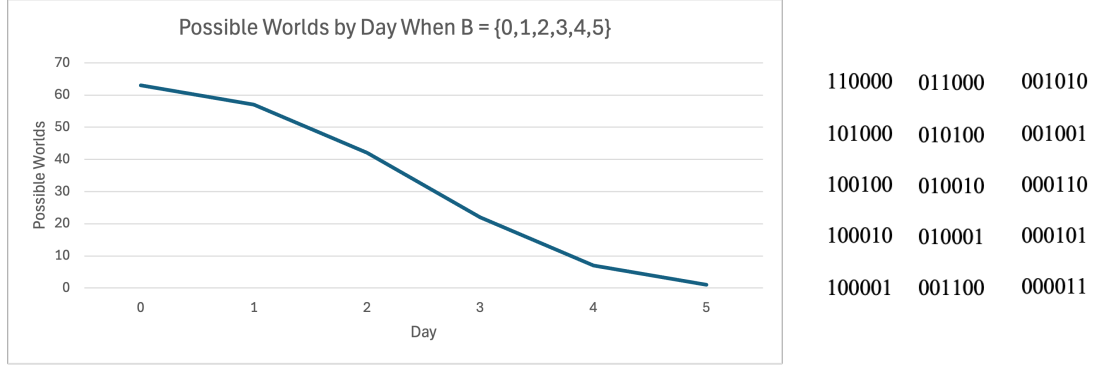


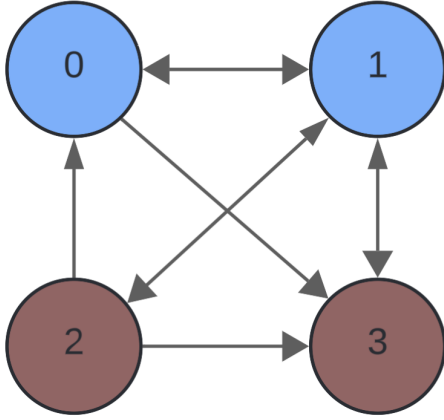
Figure 11: Left: Possible worlds left after each day of reasoning. Right: Possible worlds where there are two agents with blue eyes.

5.2 Convergence Under Partial Visibility

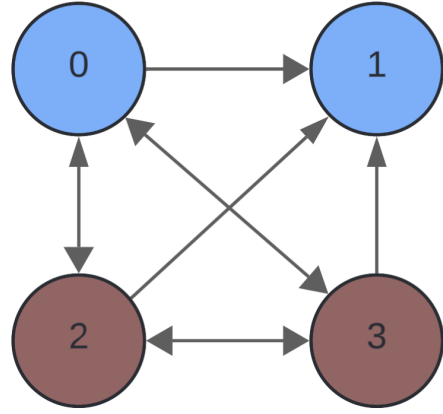
To evaluate the effects of partial visibility, we used four representative connected visibility graphs and conducted the same day-by-day update process on each of these graphs (Figure 12). We recorded the sizes of the sets of candidate worlds that remained after each day. These graphs show the rate at which public updates eliminate possible worlds. It is easy to visually distinguish between runs that eventually converge to one, and those that reach a fixed point.

In the first graph of Figure 12 (Graph A), the line depicts an ongoing decrease in the possible worlds for multiple days until it shrinks to a single world. The graph is well connected and that allows new observations to remove additional candidates from consideration. This allows the model to keep shrinking. However, in the three other graphs, the lines have a very different shape. The possible world set decreases rapidly during the initial days, but then flattens out for the remainder of the run. That is, once the possible world set has shrunk to a number higher than one, there will be no additional worlds removed by subsequent public updates, even though the visibility graph remains connected throughout the run.

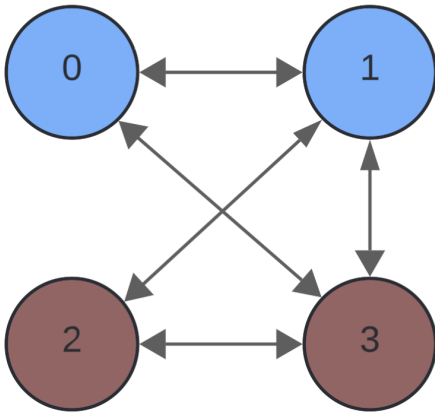
The three graphs that did not converge were chosen because they followed a commonly observed behaviour in the testing. First, when a blue-eyed agent has no visibility (Graph B), the agent's observation still has many of the candidate worlds remaining at the end of the day and the public observations made on a given day do not lead to any knowledge. Second, when visibility including a blue-eyed agent is highly asymmetric (two agents cannot see each other) (Graph C), the agents find it difficult to form the higher-order beliefs necessary to eliminate worlds. The run can not converge to one world even though the visibility graph is connected. Third, when visibility is generally sparsely distributed among the agents (Graph D), each agent's observation still has a lot of the remaining worlds and therefore each public update eliminates a relatively small portion of the candidate worlds.



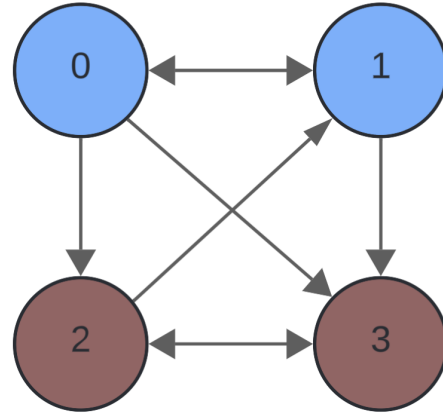
(a) Solvable.



(b) Agent one is blind.



(c) Agent 0 and 2 can't see each other.



(d) Sparse visibility.

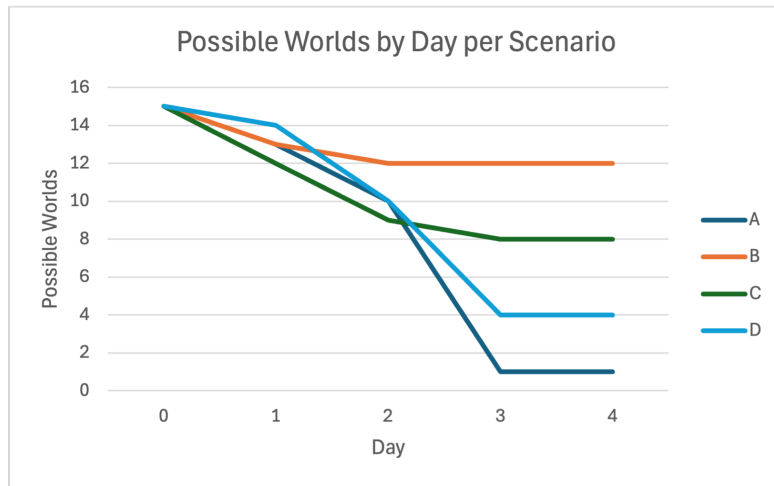


Figure 12: Connected visibility graphs with their day-by-day world elimination. A blue node means that the agent has blue eyes and an arrow pointing away from a node means it can see the agent it points towards.

5.3 Enumerating Visibility Matrices

Our enumeration was performed using two variants based on the two forms of solutions mentioned previously. The first variant used an Instance Enumerator to determine for how many visibility matrices a fixed blue-eye configuration is solvable. The second variant used a Uniform Enumerator to determine if convergence for every visibility matrix using every blue-eye configuration possible.

Uniform enumerations were extremely restricted. In every case we examined, there was only 1 uniformly solvable visibility matrix, the full visibility matrix. This is because the all-blue configuration is the only configuration that allows the elimination process to produce certainty among all agents. With partial visibility, at least one agent lacks sufficient knowledge that they can leave. Therefore, it keeps ambiguity in the all-blue configuration and prevents uniform solvability. The full visibility matrix is the only structure that supports convergence for all possible non-empty configurations.

The Instance Enumerator presents a much different view of the problem space, since it only requires the solvability for the given configuration instead of requiring that all possible placements be solvable. To visualize this we counted how many visibility matrices allowed for convergence with exactly k blue-eyed agents (Figure 13). We did this for the three different cases where there are 2, 3 and 4 total agents. This shows that while many visibility graphs provide a sufficient structure for easy problems involving fewer blue-eyed agents, as the number of blue-eyed agents increase, so does the rarity of solvability, until all agents being blue-eyed create the same bottleneck as the uniform case.

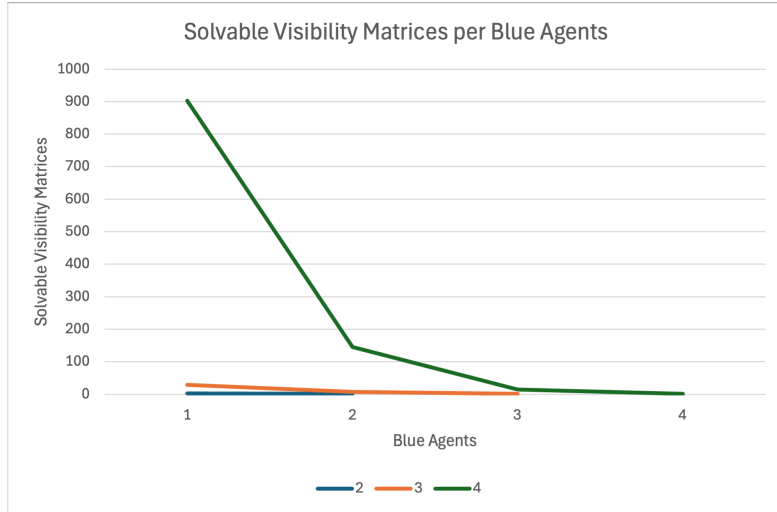


Figure 13: Amount of solvable visibility matrices (y-axis) for every amount of blue agents (x-axis) per the amount of total agents (graphs). (For clarity, you can read the highest point like this: "When there are 4 agents in a world and one of them has blue eyes, there are around 900 solvable visibility matrices")